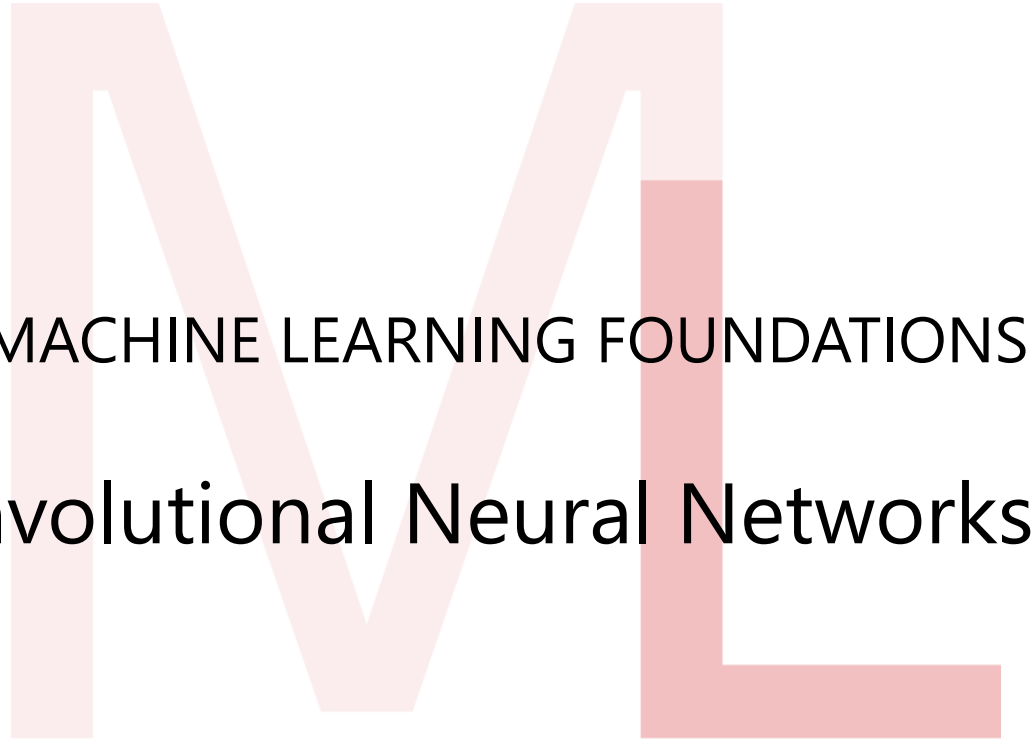
MACHINE LEARNING FOUNDATIONS

Convolutional Neural Networks (II)



University Master's Degree in

Machine Learning

upna

Universidad Pública de Navarra
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Agenda

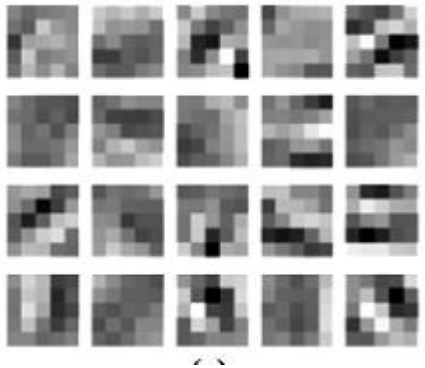
01 Recap

02 Stacking convolutions

03 LeNet5

Recap

- Convolution can be seen as the simplest feature extractor
  - The features extracted by the convolution are more complex (abstract) than pixel intensities
  - But they are still local, low-level patterns
    - Edges, lines, small textures



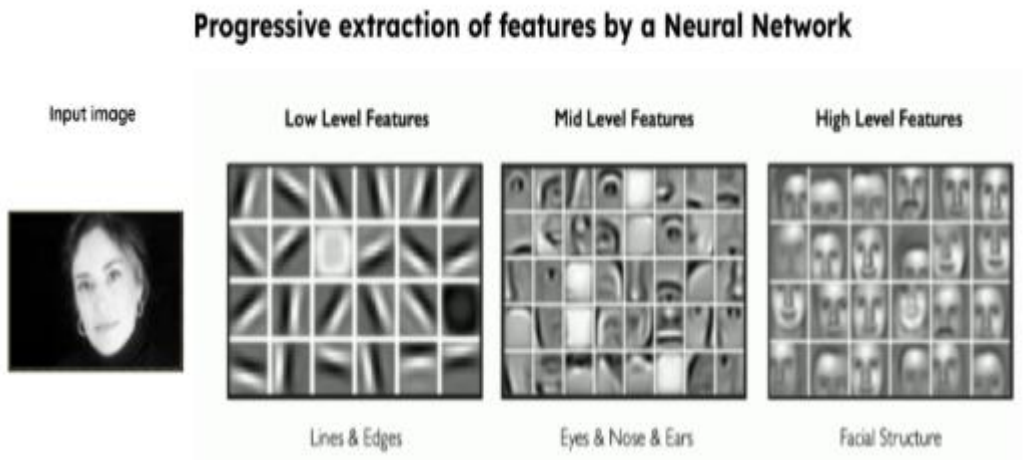
Recap

- Pooling layers allow to reduce the spatial resolution of feature maps
  - It helps controlling size and, therefore, to reduce the number of parameters
- For simple datasets, as MNIST or MNIST fashion, one single convolutional layer may be enough
  - Input -> Convolution -> Pooling -> Flatten -> MLP
- But if we go to more complex datasets, we need to detect and extract much more complex (abstract) features

Stacking convolutional layers

- In MLPs, we saw that when we go deeper, the nonlinearity of each layer produces more complex models
- Idea: if we stack more than one convolutional layer we will learn more complex patterns (features)
- This is what we call hierarchy of representations
  - If we apply convolution to the original image, we obtain a low-level feature maps
  - If we apply convolution to a low-level feature map, we obtain middle-level feature maps
  - If we continue applying convolution, we increase the abstraction of patterns

Stacking convolutional layers



LeNet

- Now, it is time to assemble everything into a fully functional CNN
- LeNet was the first published CNN
  - For recognizing handwritten digits (as in MNIST)
  - Presented in 1998!
- Main architecture:
  - 2 convolutional layers
  - 2 pooling layers
  - Flatten
  - 2 hidden layers
  - 1 output layer

Classical implementation of LeNet

- Original imagen: 32x32
- First convolutional layer: 6 convolutions size 5x5
  - Output: 6x28x28
  - Activation: Tanh
- First pooling layer: 2x2 avgpool with stride of 2
  - Output: 6x14x14
- Second convolutional layer: 16 convolutions size 5x5
  - Output: 16x10x10
  - Activation: Tanh
- Second pooling layer: 2x2 avgpool with stride of 2
  - Output: 16x5x5

Classical implementation of LeNet

- Flatten
  - Output: 400
- First hidden layer: 120 neurons
  - Output: 120
  - Activation: Tanh
- Second hidden layer: 84 neurons
  - Output: 84
  - Activation: Tanh
- Output layer: 10 neurons
  - Output: 10

